

This Was Not For Me

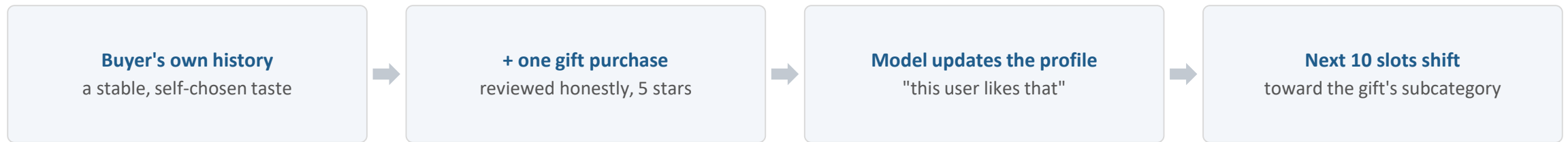
Detecting gift purchases as a distinct class of noise in e-commerce recommender systems

AI in Marketing Capstone, Group 2 — [Team Member 1] / [Team Member 2] / [Team Member 3]

github.com/edasaruhan/SIC_AI_17_Capstone_Group_2 · 22 September 2026

Every number in this deck is read from committed JSON artefacts · code version 791d9d4

A five-star review can describe a taste the buyer does not have



Illustrative schematic. No review text is reproduced anywhere in this project's outputs.

The rating is honest.

Nothing is wrong with the review itself — the buyer really did like the item.

The preference signal is not.

The item was chosen for somebody else. It is evidence about a third party.

Nothing in the data marks it.

Purchase-type is not a field. It exists only in free text, if at all.

Recommender research treats noise as accidental — mis-clicks, bots, position bias. A gift is none of those: it is a deliberate, correctly recorded purchase whose preference content belongs to someone who is not in the dataset.

The wrong signal does not stay in the model — it becomes spend

+2.0 pts

of the top-10 list is handed to the gift's subcategory

Sequential model × Toys and Games, C0 vs C1, 29,577 test users

0.66

self-purchases before that excess halves

95% CI [0.53, 0.82] · ≈2.4 weeks (approximate)

+4.9 pts

of slots go to gift-only subcategories

Classical model × Toys, waste share C0 – C1, 95% CI [4.76, 5.04] pts

WHERE THE BUDGET LEAKS

Channel	What the contaminated model does	Consequence
On-site recommendations	Reserves slots for a subcategory the customer never buys for themselves	Lower slot productivity; the customer's real next purchase is pushed out
Retargeting / email	Rebuilds the audience around the gift's category	Impressions billed against an interest that does not exist
Lifecycle timing	Reads a December gift peak as a taste shift	Seasonal campaign aimed at the wrong segment

Four questions, fixed before the experiment was run

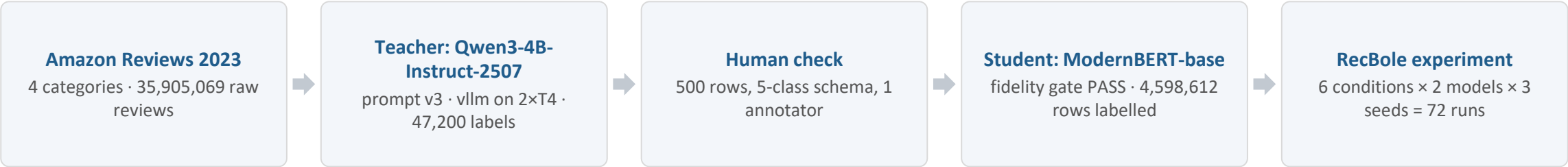
Question		How it is answered
RQ1	How common are gift purchases, and when?	Human-calibrated prevalence per category + month-of-year profile
RQ2	Does removing gift rows improve prediction of the user's own next purchase?	C1 vs a placebo (C4) that deletes the same NUMBER of random rows
RQ3	Is it better to delete gifts or to flag them?	C3 (shadow token) against both C1 (delete) and C0 (do nothing)
RQ4	How long does a recommendation list stay contaminated?	M2 half-life in self-purchases, M1 wasted slot share

The interpretation rule was written down BEFORE any result was seen

A 95 % bootstrap interval that excludes zero in the expected direction → reported as a LOWER BOUND (alt_sınır), because label noise pulls a placebo contrast toward the placebo (assuming the detector's mistakes are no more informative than random rows — A4). An interval containing zero → NOT DETECTABLE (saptanamadı) — never "no effect". An interval excluding zero in the opposite direction → NEGATIVE (negatif), reported as such.

Dated in repo/docs/DECISIONS.md. Thresholds were never edited after a result was seen; the 500 human-labelled rows were never used for tuning.

One pipeline: a big model teaches a small one, then the small one labels everything

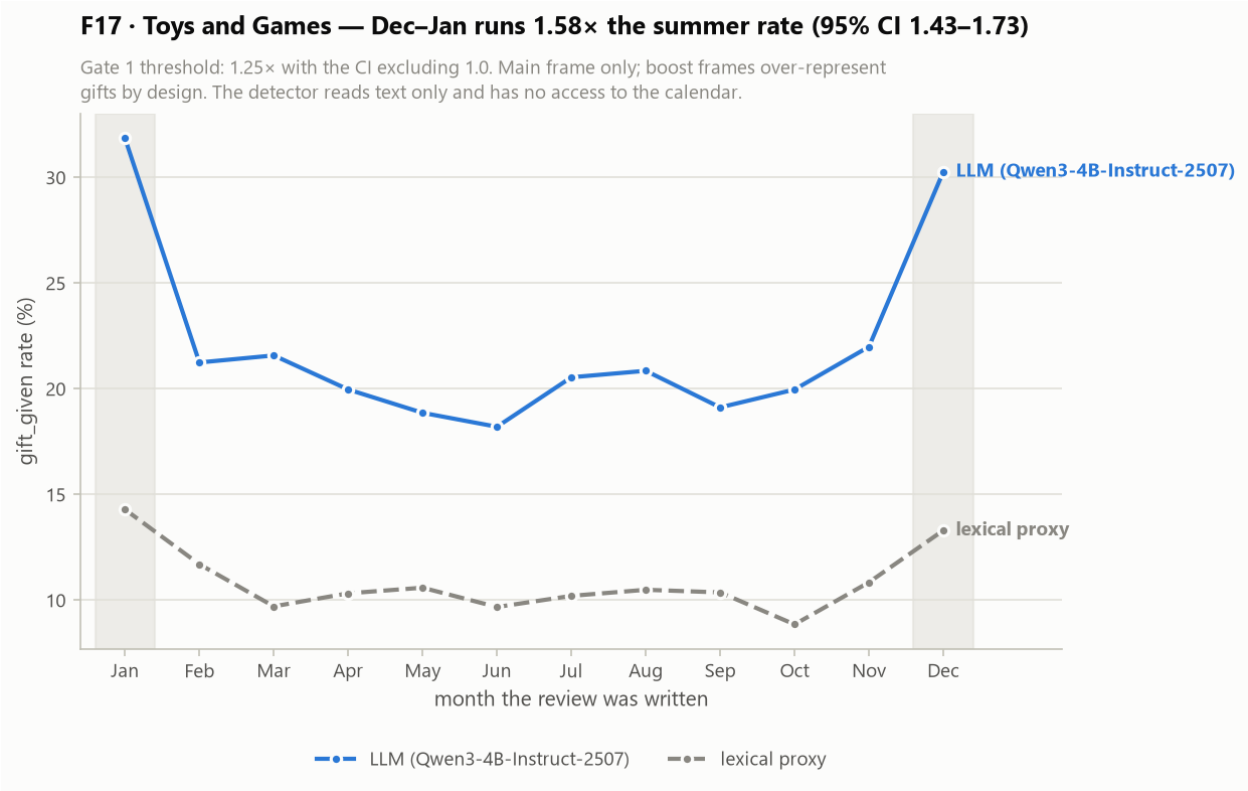


FROM RAW REVIEWS TO A TRAINABLE CORPUS

Category	Raw reviews	After 5-core	Role in the study
Toys and Games	16,260,406	2,164,018	High-gift treatment category
Grocery and Gourmet Food	14,318,520	2,434,594	Low-gift control category
Video Games	4,624,615	368,476	Detector validation only
All Beauty	701,528	0	Detector only — 5-core is empty, so no experiment is possible

Filters, in order: verified purchase → at least 5 words of text → de-duplication → 5-core on users and items. Only the two experiment categories were labelled at full scale; the distilled student exists because labelling 4.6 M rows with the teacher was not affordable.

The detector had to earn its use — four criteria, all fixed in advance



F17 · Toys and Games: LLM label rate vs the lexical keyword proxy, by month

GATE 1 — ALL FOUR CATEGORIES

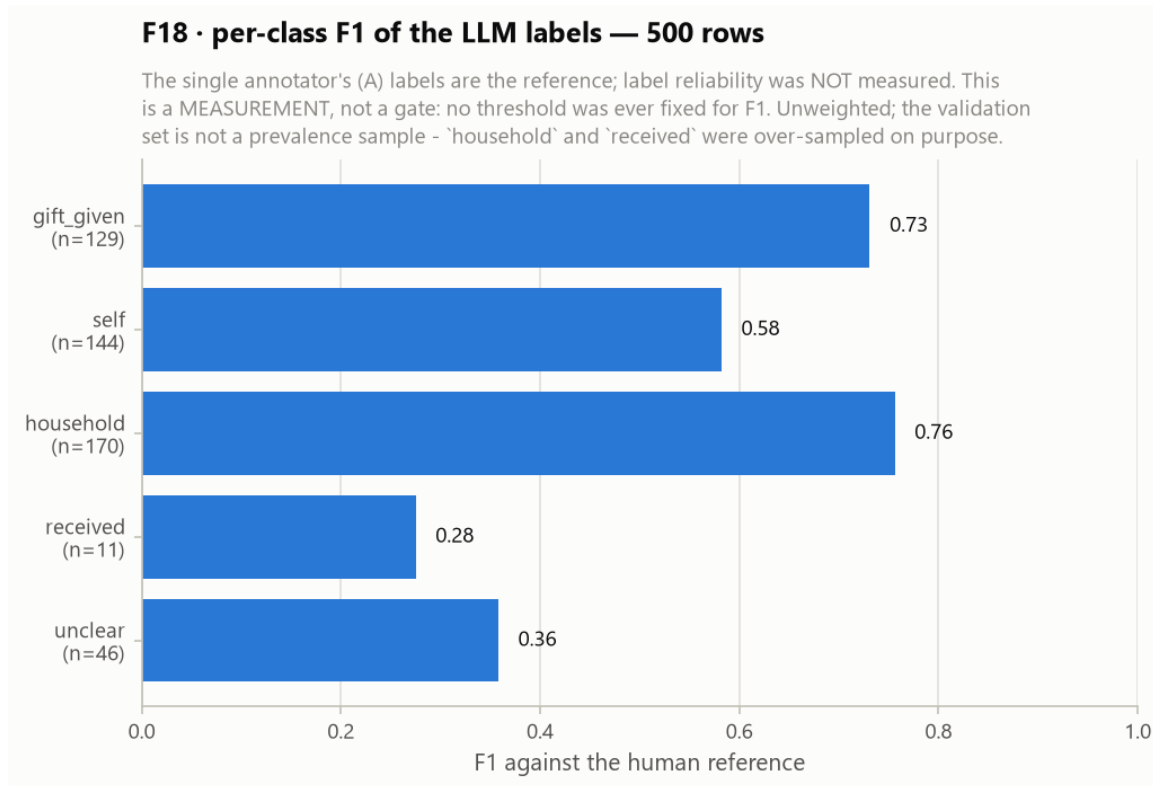
Criterion	Threshold	Toys
1 · December–January vs summer	≥ 1.25×	1.58×
2 · Found beyond the keyword proxy	≥ 1%	18.0%
3 · Schema health (parse, class mix)	no failure	pass
4 · 200-row trial agreement	≥ 0.7	0.835

VERDICT PER CATEGORY

Toys & Games	PASS 4/4
Grocery & Gourmet Food	PASS 4/4
Video Games	PASS 4/4
All Beauty	PASS 4/4

Seasonality is the criterion a keyword list cannot fake: it is a property of the calendar, not of the wording.

The detector did not meet its own accuracy thresholds. We report that first.



Two pre-registered gates were MISSED

macro-F1 $\geq 0.75 \rightarrow$ measured 0.5405 [0.4897, 0.5869]

gift precision $\geq 0.80 \rightarrow$ measured 0.68 (recall 0.79, F1 0.7312)

Week-4 verdict recorded in the artefact: INCOMPLETE

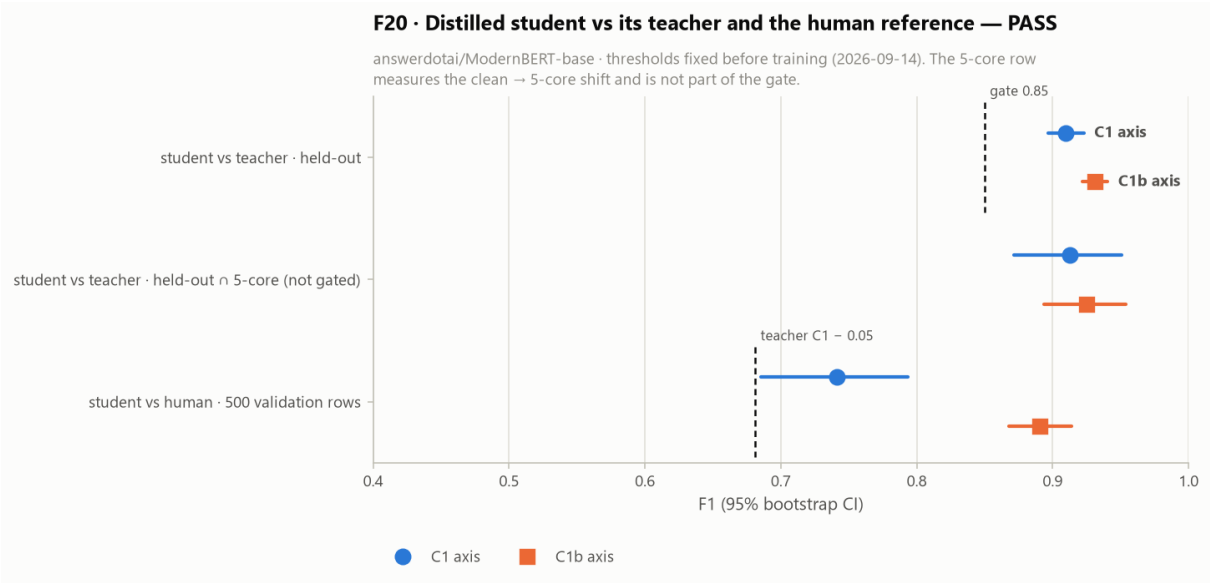
Label reliability was never measured

Planned: 3 annotators, Fleiss kappa ≥ 0.6 . Actual: 1 annotator, so no agreement statistic exists.

Why the study is still readable

The placebo holds the amount of deleted data fixed, and the effect scales with the gift share — neither depends on detector accuracy. What the weak detector changes is what is measured: the rows it FLAGS as gifts.

A distilled student reproduced the teacher's labels closely enough to scale



F20 · student vs teacher on the held-out split

FIDELITY GATE — 3 CRITERIA, FIXED BEFORE TRAINING

Criterion	Threshold	Measured
F1 vs teacher, narrow axis (C1)	≥ 0.85	0.9098
F1 vs teacher, broad axis (C1b)	≥ 0.85	0.9313
Not worse than teacher vs human	drop ≤ 0.05	0.7415 vs 0.7312

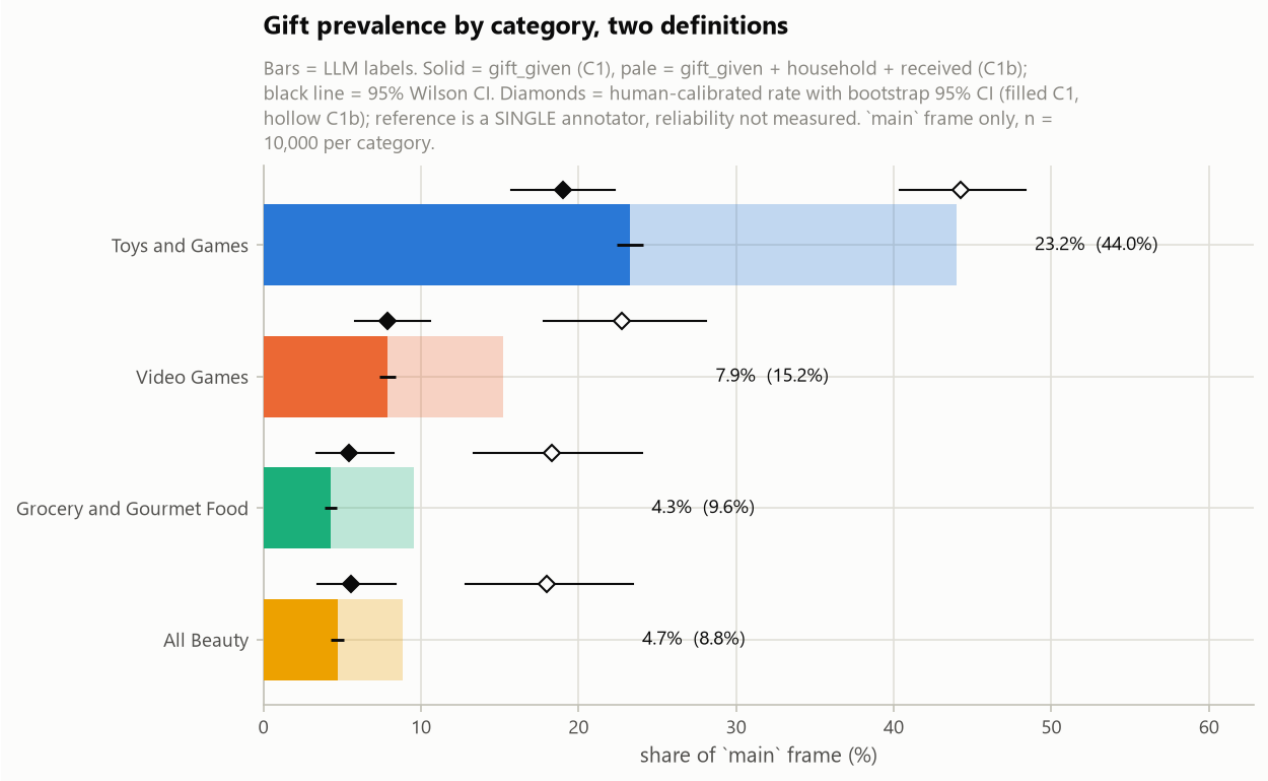
GATE PASS · 3/3

What this does and does not say

Trained on 41,988 teacher labels, checked on 4,667 held-out rows, then run over 4,598,612 interactions (2,164,018 Toys + 2,434,594 Grocery).

The gate measures agreement with the TEACHER, not correctness. The student inherits the teacher's errors — which is exactly why the accuracy limits on the previous slide still apply at 4.6 M rows.

RQ1 · About one in five Toys reviews is a gift — and every category peaks in December



F19 · human-calibrated prevalence, narrow (gift_given) axis

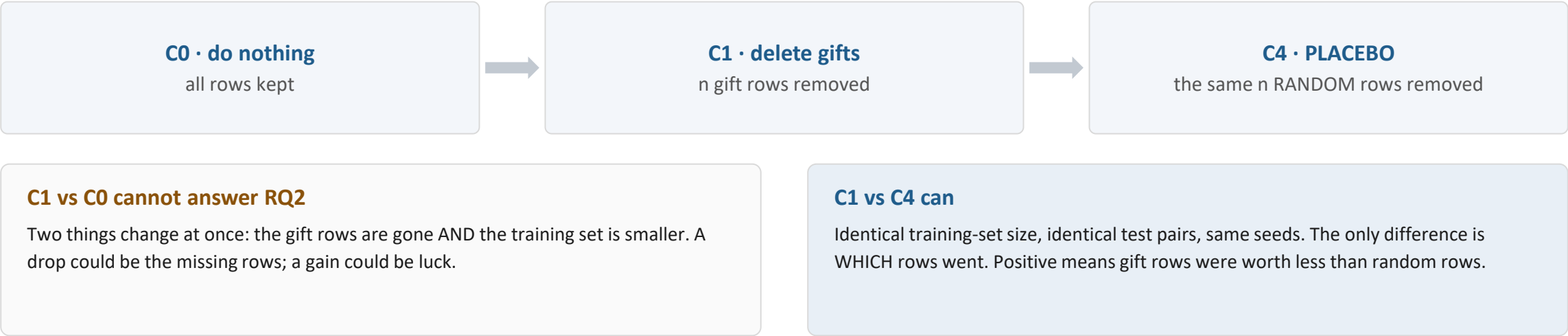
Category	Calibrated	95 % CI	Raw LLM
Toys & Games	19.0%	[15.8, 22.3]	23.2%
Grocery	5.4%	[3.4, 8.3]	4.3%
Video Games	7.9%	[5.8, 10.6]	7.9%
All Beauty	5.5%	[3.4, 8.4]	4.7%

DECEMBER–JANUARY ÷ SUMMER

Toys & Games 1.58× [1.43, 1.73]
Grocery 1.71× [1.33, 2.23]
Video Games 2.08× [1.74, 2.49]
All Beauty 2.06× [1.61, 2.63]

Calibration maps the LLM's raw rate through the confusion matrix measured on the 500 human rows, so the interval carries the detector's error — this is why Toys moves from 23.2 % to 19.0 %.

Deleting gifts also deletes data — so the comparison must hold the data loss fixed



ALL SIX CONDITIONS

C0	C1	C4	C1b	C4b	C3
baseline nothing removed	narrow gift_given removed	placebo for C1 same n, random	broad + household, received	placebo for C1b same n, random	flag, don't delete shadow token

C2 (down-weighting) was specified but not run: RecBole's SASRec loss does not accept per-interaction weights without patching the library, and patching it would have broken the "same code for every condition" rule.

Gate 2 · the setup is valid in all four cells, so the contrasts mean something

Criterion	What it rules out	Toys · SASRec	Toys · BPR	Grocery · SASRec	Grocery · BPR
1 · Every run scored on the same test pairs	Comparing different test sets	PASS	PASS	PASS	PASS
2 · Each condition's item universe $\subseteq$ C0's	A condition inventing items	PASS	PASS	PASS	PASS
3 · Neither placebo is BETTER than C0	Deletion helping by itself	PASS	PASS	PASS	PASS
4 · C0 seed-to-seed CV < 0.1	Runs too unstable to compare	PASS · 0.012	PASS · 0.025	PASS · 0.005	PASS · 0.002

CRITERION 3 IN NUMBERS — NEITHER PLACEBO IS BETTER THAN C0 (THE CI'S UPPER END MAY TOUCH ZERO, BUT NEVER RISES ABOVE IT)

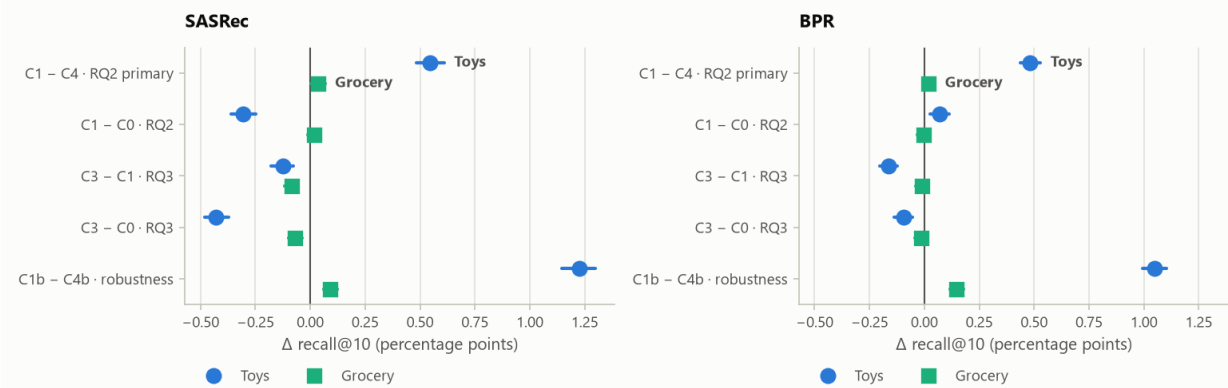
Cell	C4 – C0, Recall@10	C4b – C0, Recall@10
Toys · SASRec	-0.008510 [-0.0092, -0.0079]	-0.022462 [-0.0234, -0.0216]
Toys · BPR	-0.004109 [-0.0046, -0.0036]	-0.010853 [-0.0114, -0.0103]
Grocery · SASRec	-0.000201 [-0.0005, 0.0001]	-0.001638 [-0.0020, -0.0013]
Grocery · BPR	-0.000235 [-0.0005, 0.0000]	-0.001357 [-0.0017, -0.0011]

18 runs per cell, 72 in total. Criterion 1 is enforced by a SHA-256 hash of the test pairs written into every run report — one distinct hash per cell, checked by the gate rather than assumed.

RQ2 · Removing gifts beats removing the same number of random rows

F21 · Next-purchase accuracy when gifts are removed (C1) or flagged (C3)

Per-user differences averaged over seeds; lines are 95% paired-bootstrap CIs. An interval covering zero on C1 – C4 reads as “not detectable at this label precision”, not “no effect”.



F21 · per-cell contrasts with 95 % paired-bootstrap intervals

C1 – C4, RECALL@10 (PRIMARY)

Cell	Δ	95 % CI	Rel.	Rule
Toys · SASRec	+0.005449	[0.0048, 0.0061]	+15.7%	lower bound
Toys · BPR	+0.004822	[0.0044, 0.0053]	+40.0%	lower bound
Grocery · SASRec	+0.000360	[0.0000, 0.0007]	+1.2%	lower bound
Grocery · BPR	+0.000201	[-0.0001, 0.0005]	+1.4%	not detectable

DOSE–RESPONSE: TOYS – GROCERY

Model	Axis	Δ	95 % CI
SASRec	C1-C4	+0.005089	[0.0044, 0.0058]
SASRec	C1b-C4b	+0.011330	[0.0106, 0.0121]
BPR	C1-C4	+0.004621	[0.0041, 0.0052]
BPR	C1b-C4b	+0.009001	[0.0084, 0.0096]

All four dose–response intervals exclude zero: the effect is larger where gifts are more common. That is the pattern a real mechanism produces.

But "better than the placebo" is not "better than doing nothing"

The placebo contrast isolates the mechanism. The C1 – C0 contrast is what a practitioner would actually feel.

Cell	C1 – C4 (vs placebo)	Rule	C1 – C0 (vs doing nothing)	Rule	C1b – C4b (broad axis)	Rule
Toys · SASRec	+0.005449 +15.7%	lower bound	-0.003061 -7.1%	negative	+0.012253 +59.3%	lower bound
Toys · BPR	+0.004822 +40.0%	lower bound	+0.000713 +4.4%	positive	+0.010473 +197.4%	lower bound
Grocery · SASRec	+0.000360 +1.2%	lower bound	+0.000159 +0.5%	CI includes zero	+0.000923 +3.2%	lower bound
Grocery · BPR	+0.000201 +1.4%	not detectable	-0.000034 -0.2%	CI includes zero	+0.001472 +11.2%	lower bound

The mechanism is real

C1 – C4 excludes zero in three of four cells, and the dose–response holds. Gift rows carry less usable preference signal than random rows.

The net gain is model-dependent

On Toys, deleting gifts outright helps BPR (+4.4%) but hurts SASRec (-7.1%) — the sequence model loses context with the rows.

Widening the definition widens the gap

C1b – C4b is larger than C1 – C4 in all four cells, which is consistent with household and received purchases carrying the same kind of noise.

RQ3 • Flagging with a shadow token was worse than both alternatives

C3 keeps every gift row but replaces the item with a per-subcategory shadow token, so the model can learn "something happened here" without learning the item.

Cell	C3 – C1 (flag vs delete)	Rule	C3 – C0 (flag vs nothing)	Rule
Toys · SASRec	-0.001235 [-0.0018, -0.0008]	negative	-0.004296 [-0.0048, -0.0037]	negative
Toys · BPR	-0.001647 [-0.0020, -0.0012]	negative	-0.000934 [-0.0014, -0.0005]	negative
Grocery · SASRec	-0.000850 [-0.0012, -0.0005]	negative	-0.000691 [-0.0010, -0.0004]	negative
Grocery · BPR	-0.000102 [-0.0004, 0.0002]	CI includes zero	-0.000135 [-0.0004, 0.0001]	CI includes zero

What we found

Negative against deletion in three of four cells, and against doing nothing in three of four. The shadow token adds vocabulary and sequence positions without adding information the model can use.

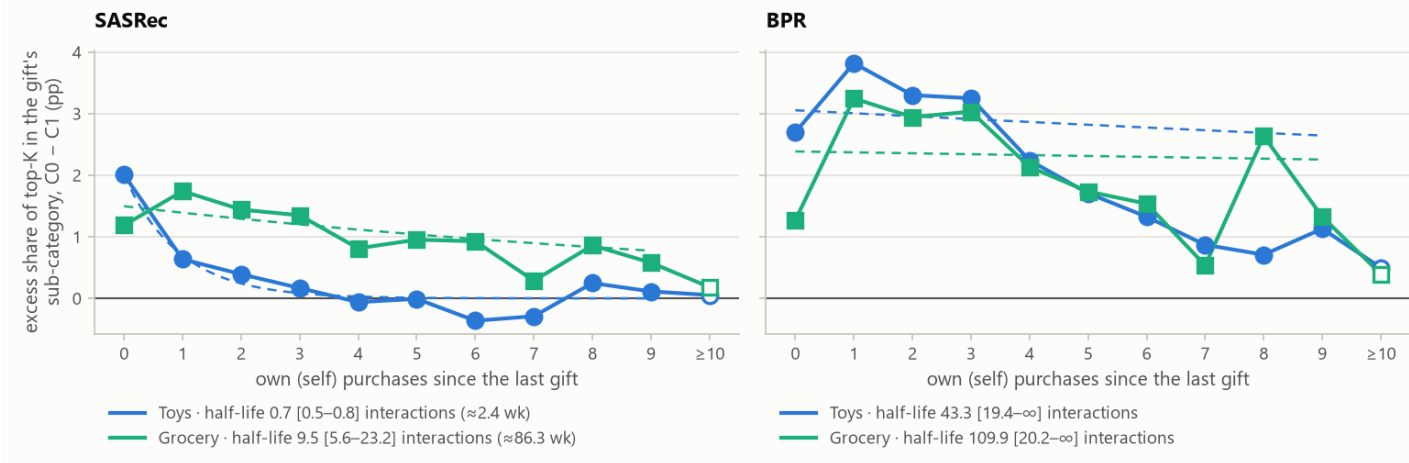
What we are NOT claiming

That flagging is a bad idea in general. One implementation was tested: a per-subcategory shadow token in the item sequence. Type-aware embeddings, a separate purchase-type channel, or loss re-weighting were not tested.

RQ4 · The excess halves after about one self-purchase — roughly two to four weeks

F22 · M2 — how long a gift keeps pulling recommendations toward its sub-category

Bucket means; dashed = $A \cdot \exp(-\lambda n)$ fit on filled buckets (hollow: too few users or the pooled $\geq$ bucket; gaps: no users). Weeks use the median gap between reviews, which lag purchases. BPR ignores order, so no decay is expected there.



0.66

self-purchase half-life

95 % CI [0.53, 0.82]

≈2.4 wk

approximate calendar time

review date ≠ purchase date

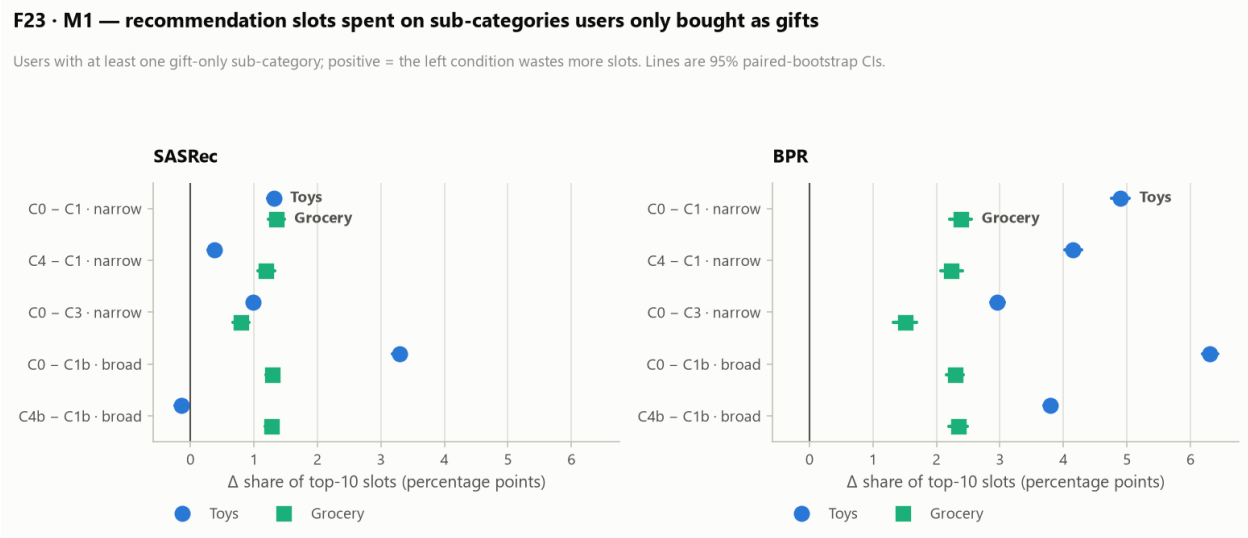
Read it like this

A gift opens +2.0 pts of extra top-10 share for its subcategory. After one purchase the user makes for themselves, less than half of that is left; after four it is gone.

BPR shows decay too, but the pre-registered fit gives no finite half-life: the gift is DILUTED by later purchases rather than forgotten, because BPR never sees order.

F22 · excess top-10 share for the gift's subcategory vs the number of self-purchases after the gift, with the pre-registered exponential fit

RQ4 · In the classical model, roughly one slot in twenty goes to gift-only interest



F23 · share of top-10 slots drawn from subcategories the user entered only as a gift

Cell	C0 – C1	C4 – C1 (net of data loss)
Toys · SASRec	+1.3 pts [1.22, 1.41]	+0.4 pts [0.28, 0.47]
Toys · BPR	+4.9 pts [4.76, 5.04]	+4.2 pts [4.02, 4.29]
Grocery · SASRec	+1.4 pts [1.23, 1.48]	+1.2 pts [1.06, 1.33]
Grocery · BPR	+2.4 pts [2.21, 2.56]	+2.2 pts [2.07, 2.41]

What the numbers say

On Toys, 32% of test users have a subcategory they entered only as a gift. The classical model gives 16.3% of their top-10 to it; the gift-free model gives 11.4%.

Holding data loss fixed (C4 – C1) keeps most of that gap in the classical model but almost none of it in the sequential one.

"Waste" is an upper bound by construction: we did not measure whether those slots convert. A customer may later buy in that subcategory for themselves.

What we claim, what we do not, and what comes next

We claim

- Gift purchases are measurable at scale, and common: 19 % of Toys reviews.
- Gift rows carry less usable preference signal than random rows, with a dose–response across two categories.
- A gift's excess recommendation share halves after about one self-purchase in the sequential model.
- The one flagging method we tested is worse than deleting.

We do not claim

- That the detector is accurate: macro-F1 0.54 against one annotator, both pre-registered gates missed.
- That deleting gifts is a recommended production change — the net effect against doing nothing is model-dependent.
- That this generalises beyond two Amazon categories, two models and English-language reviews.
- That wasted slots are lost revenue; conversion was never measured.

Next, in order of value

- A second and third annotator on the same 500 rows, and the Fleiss κ the design asked for.
- Purchase-type as a model input instead of a deletion rule — the C2 that RecBole would not run.
- A held-out category the detector never saw during prompt development.
- Conversion-linked data, so "waste" becomes a measured cost.

Everything on these slides is reproducible from the repository

`python repo/scripts/demo.py` — prints every headline number above from the committed JSON artefacts, with no data files and no dependencies beyond the standard library.
`github.com/edasaruhan/SIC_AI_17_Capstone_Group_2` code version 791d9d4

A1 · The six conditions, exactly as they were run

Condition	Definition	Rows removed	Purpose
C0	Baseline — the full 5-core training set	none	Reference for every other condition
C1	Narrow: rows labelled gift_given are deleted from TRAINING	n_gift	RQ2 treatment
C4	Placebo for C1 — the same NUMBER of rows, chosen at random	n_gift	Separates "which rows" from "how many rows"
C1b	Broad: gift_given + household + received deleted	n_broad	Sensitivity to the label definition
C4b	Placebo for C1b	n_broad	Same, for the broad axis
C3	Gift rows kept; item replaced by a per-subcategory shadow token	none	RQ3 — flag instead of delete

Only training rows are touched; validation and test rows are identical in every condition, which is what makes the runs comparable — and is also a limitation: a gift sitting in a user's validation row is never removed, so the sequential model still sees it as the last item of the test input. Placebo row selection uses a per-condition seed derived as $(\text{base_seed} + \text{zlib.crc32}(\text{condition_name})) \bmod 2^{31}-1$, never Python's hash(), which is randomised per process.

A2 · Both gates in full, with the numbers that decided them

GATE 1 — MAY THE DETECTOR BE USED? (PER CATEGORY)

Criterion	Threshold	Toys	Grocery	Video Games	All Beauty
1 · Dec–Jan ÷ summer ratio	≥ 1.25×	1.58×	1.71×	2.08×	2.06×
2 · Rate found beyond keyword proxy	≥ 1%	18.0%	3.1%	4.9%	3.3%
3 · Schema health	no failure	pass	pass	pass	pass
4 · 200-row trial agreement	≥ 0.7	0.835	0.835	0.835	0.835
Verdict		PASS 4/4	PASS 4/4	PASS 4/4	PASS 4/4

GATE 2 — MAY THE CONTRASTS BE INTERPRETED? (PER CATEGORY × MODEL CELL)

Criterion	Mechanism	Toys · SASRec	Toys · BPR	Grocery · SASRec	Grocery · BPR
1 · Same test pairs everywhere	SHA-256 of the test pairs, written per run	PASS	PASS	PASS	PASS
2 · Item universe ⊆ C0	Set comparison against C0's item table	PASS	PASS	PASS	PASS
3 · Neither placebo is better than C0	Paired bootstrap; the CI's lower end must not sit above zero	PASS	PASS	PASS	PASS
4 · C0 seed-to-seed CV < 0.1	Std ÷ mean of C0's recall@10 over the three seeds	0.012	0.025	0.005	0.002

Gate 2 checks 18 runs per cell (72 in total) and is re-evaluated from the run reports, not from a cached verdict.

A3 · Every contrast, every cell, Recall@10 with 95 % intervals

Contrast	Meaning	Toys · SASRec	Toys · BPR	Grocery · SASRec	Grocery · BPR
C1 – C4	Delete gifts vs delete the same number at random	+0.005449 [0.0048, 0.0061] lower bound	+0.004822 [0.0044, 0.0053] lower bound	+0.000360 [0.0000, 0.0007] lower bound	+0.000201 [-0.0001, 0.0005] not detectable
C1b – C4b	Same, broad label axis	+0.012253 [0.0114, 0.0130] lower bound	+0.010473 [0.0099, 0.0110] lower bound	+0.000923 [0.0006, 0.0012] lower bound	+0.001472 [0.0012, 0.0018] lower bound
C1 – C0	Delete gifts vs do nothing	-0.003061 [-0.0036, -0.0025] negative	+0.000713 [0.0003, 0.0011] positive	+0.000159 [-0.0001, 0.0005] CI includes zero	-0.000034 [-0.0003, 0.0003] CI includes zero
C3 – C1	Flag vs delete	-0.001235 [-0.0018, -0.0008] negative	-0.001647 [-0.0020, -0.0012] negative	-0.000850 [-0.0012, -0.0005] negative	-0.000102 [-0.0004, 0.0002] CI includes zero
C3 – C0	Flag vs do nothing	-0.004296 [-0.0048, -0.0037] negative	-0.000934 [-0.0014, -0.0005] negative	-0.000691 [-0.0010, -0.0004] negative	-0.000135 [-0.0004, 0.0001] CI includes zero

Paired bootstrap over per-user Recall@10, averaged across seeds 42, 1337, 2024 before resampling; 117,386 test users in Toys, 238,756 in Grocery. Recall@10 replaced NDCG@10 as the primary metric before the runs, because a contaminated list is a coverage problem before it is a ranking problem — the change is dated in DECISIONS.md and NDCG is still reported alongside.

A4 · The limitations we would raise ourselves

#	Limitation	What it costs the conclusion
1	Two pre-registered detector gates were missed (macro-F1 0.5405 vs ≥ 0.75 ; gift precision 0.68 vs ≥ 0.80)	Absolute prevalence is uncertain; effects measure detector-flagged rows
2	The primary metric changed from NDCG@10 to Recall@10 before the runs	Pre-registered, dated, and NDCG is still reported — but it is a change
3	1 human annotator; the planned Fleiss κ was never measured	No reliability estimate for the reference labels; a September retry came back AI-assisted and was discarded
4	C1 deletes rows the detector believes are gifts, including its false positives	The treatment is "delete what the detector flags", not "delete gifts"
5	Gifts in a user's validation row are never removed	Direction of the effect on the sequential model is untested
6	Epochs trained were not recorded for the 72 original runs	Seed-42 re-run: 24/24 bit-identical, 21–77 epochs, none at the 300 cap
7	Intervals resample users, not training runs (seed variance excluded)	Toys: every seed agrees in sign · Grocery: seed spread $\approx$ interval width
8	The placebo matches the number of rows, not the users they come from	Post-hoc: the Toys effect holds even for users neither condition touched
9	"Lower bound" assumes false positives are as informative as random rows	Most are purchases for the buyer's own child — not guaranteed

Ten further limitations are listed in `repo/docs/SONUCLAR.md §7`, including: two categories and two models only; English-language reviews; review date used as a proxy for purchase date; "waste" measured as slot share rather than conversion; M2 is observational and its $n = 0$ bucket is structurally different from the others (a post-hoc sensitivity check is reported alongside the pre-registered fit); subcategory taken from item metadata, which is itself incomplete.

A5 · How to check any number on any slide

Start here

- `repo/scripts/demo.py` — prints every headline number from the committed JSONs. Standard library only; no data files needed.
- `repo/docs/SONUCLAR.md` — every result with its interval and its limitation.
- `repo/docs/DECISIONS.md` — every threshold and definition, with the date it was fixed.
- `final-report/final-report.md` — the written report, with the source JSON named in each section.

Provenance stamped into every artefact

- `prevalence.json` — `code_version` 05df87b
- `gate2.json` — `code_version` 791d9d4
- `experiment_stats.json` — `code_version` 791d9d4
- `marketing_metrics.json` — `code_version` 3058448

A "-dirty" suffix means the working tree had uncommitted CODE changes when the artefact was written. Runs also carry `label_source`, `reportable` and a SHA-256 of the test pairs.

WHAT IS AND IS NOT IN THE REPOSITORY

Committed

Code, configs, 463 tests, every result JSON, every figure, the report and this deck's generator

Deliberately not committed

Review data and model weights (Kaggle datasets are private), the 500 human-labelled rows, any verbatim review text, any personal data

Two virtual environments: `.venv` for the pipeline (polars 1.44, numpy 2.5) and `.venv-recbole` for the experiment (numpy < 2, because RecBole 1.2.0 still uses `np.float_`). Slides were generated by `presentation/build_deck.py`, which reads the same JSONs — so the deck cannot drift from the artefacts. The 24 seed-42 cells were re-run with the current code: 24/24 bit-identical (`reproduction_seed42.json`).